TEXORA · LECTURE SHEET

Introduction to Computers

What a computer is and why it is so powerful, the functional units and block diagram, five generations, classifications by data handling and size, number systems with conversions, and how data is measured and used — including in the textile industry.

CSE-2105 · FUNDAMENTALS OF COMPUTING · TOPIC 1 · LEVEL 2 TERM 1

1. What is a computer?

A **computer** is an electronic machine that accepts data as input, processes it according to a stored set of instructions called a **program**, and produces useful information as output. The word comes from the Latin *computare*, to calculate, and early machines were indeed fast calculators. A modern computer is far more general: the same hardware can write a letter, design a fabric pattern, control a loom, or analyse laboratory results, simply by running a different program. This stored-program idea, described by John von Neumann in the 1940s, is the foundation of almost every computer in use today: both data and instructions live in memory, and the processor fetches and executes them one after another.

Three terms must be kept distinct. **Data** are raw, unorganised facts — numbers, names, measurements. **Information** is data that has been processed into a meaningful, useful form. A **program** is the ordered set of instructions that tells the machine how to transform the first into the second. Input, processing, output, and storage together form the **information processing cycle**, repeated billions of times per second in a modern machine.

2. Characteristics of a computer

Six characteristics explain why computers have spread into every industry, from banking to textile mills.

CHARACTERISTIC	MEANING
Speed	Operations are measured in milliseconds, microseconds, nanoseconds, and even picoseconds. A current processor executes several billion instructions per second, so a task that would take a person weeks of arithmetic is finished in a fraction of a second.
Accuracy	Given correct input and a correct program, the result is correct. Errors in computing are almost always caused by human mistakes in data or instructions — a principle

	summed up as GIGO: Garbage In, Garbage Out. The machine itself does not make arithmetic slips.
Diligence	A computer does not tire, lose concentration, or become bored. It performs the millionth repetition with exactly the same speed and accuracy as the first, which makes it ideal for repetitive work such as payroll calculation or loom monitoring.
Versatility	The same machine performs very different jobs by changing the program. Versatility is limited only by the programs written for it and by its input and output devices.
Storage	Large volumes of data and programs can be stored in a small physical space and retrieved instantly and exactly when needed. Storage is organised in a hierarchy from fast, small registers to vast, slow archives — the subject of Topic 2.
Automation	Once a program is loaded, work proceeds automatically without step-by-step human intervention. The user supplies the program and the data; the machine sequences the rest.

Two limitations balance this picture. A computer has no intelligence of its own: it cannot think, judge, or act beyond its instructions, and it has no feelings or common sense. Every decision it appears to make was anticipated, in some form, by a programmer.

3. Block diagram and functional units

Every general-purpose computer, from a phone to a supercomputer, contains the same four functional units: an input unit, a central processing unit, an output unit, and storage. The central processing unit itself contains the arithmetic and logic unit, the control unit, and the primary memory unit.

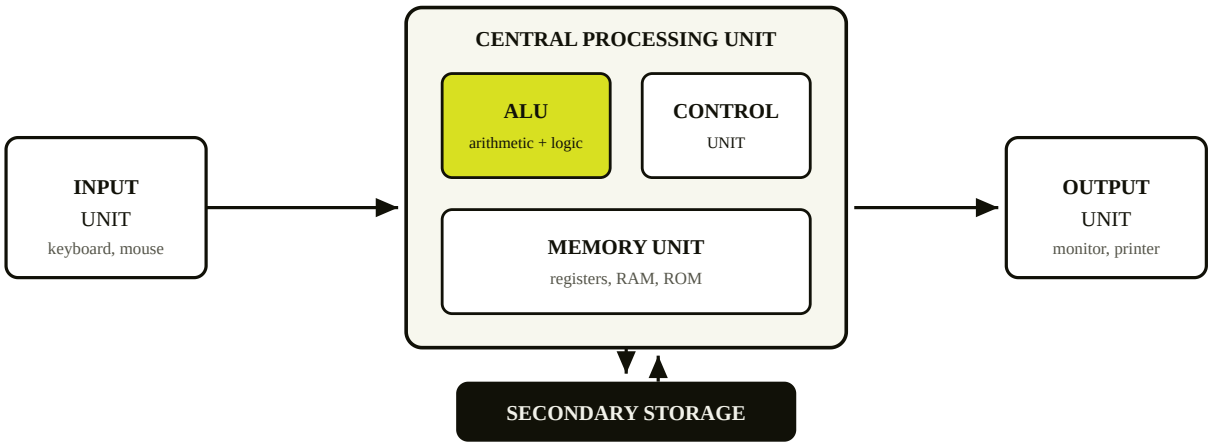


Figure 1 — Block diagram of a computer: input, CPU (ALU, control unit, memory unit), output, and secondary storage

Input unit. Accepts data and instructions from the outside world and converts them into the binary form the machine uses internally. Examples are the keyboard, mouse, scanner, microphone, and

barcode reader. Input is not simply typing: a temperature sensor feeding a dyeing machine controller is also an input device, because it supplies data to the processor.

Arithmetic and Logic Unit (ALU). Performs all calculations (addition, subtraction, multiplication, division) and all logical operations and comparisons (AND, OR, NOT, greater than, equal to). Results are held briefly in small, very fast storage locations inside the processor called registers before being sent back to memory.

Control Unit (CU). The coordinator of the whole system. It fetches each instruction from memory, decodes what the instruction means, and sends timed control signals to the ALU, memory, and input and output devices so that the right operation happens in the right order. It does not process data itself; it directs the processing, rather as a traffic controller directs vehicles without carrying passengers.

Memory unit. Holds the data to be processed, the intermediate results, the final results waiting for output, and the program being executed. Primary memory is fast and directly accessible to the CPU but limited in size; secondary storage holds far more, permanently, at lower speed.

Output unit. Converts the binary results of processing back into a human-usable form: characters on a monitor, print on paper, sound, or a control signal to a machine. Together, input and output devices are called peripherals.

4. Generations of computers

Computer history is conventionally divided into five generations, each defined by its main electronic technology. Each generation was smaller, faster, cheaper, and more reliable than the one before.

GENERATION	PERIOD	KEY TECHNOLOGY	FEATURES AND EXAMPLES
First	1940–1956	Vacuum tubes	Huge, hot, power-hungry machines; punched cards for input; machine language only; very frequent tube failures. Examples: ENIAC, UNIVAC I.
Second	1956–1963	Transistors	Smaller, cooler, faster and more reliable than tubes; magnetic core memory; assembly language and early high-level languages such as FORTRAN and COBOL; batch processing. Example: IBM 7094.

Third	1964–1971	Integrated circuits (IC)	Many transistors fabricated on one silicon chip; keyboards and monitors replace punched cards; operating systems and time-sharing appear. Example: IBM System/360.
Fourth	1971–present	Microprocessor, VLSI	The whole CPU on a single chip; Very Large Scale Integration packs hundreds of thousands, later millions, of transistors per chip; personal computers, graphical interfaces, and networks spread everywhere. Example: Intel 4004 (1971) and all modern PCs.
Fifth	Present and beyond	AI, ULSI	Ultra Large Scale Integration, parallel processing, and artificial intelligence: natural language understanding, pattern recognition, expert systems, and machine learning. Still developing.

The pattern across generations is consistent: integration increases, so cost and size fall while speed and reliability rise. A single modern phone chip contains more transistors than entire rooms of first-generation equipment.

5. Classification of computers

5.1 By data handling

TYPE	HOW IT WORKS	EXAMPLES AND USES
Analog	Works on continuous physical quantities such as voltage, pressure, or shaft rotation. It measures rather than counts, so results are approximate.	Slide rule, analog thermometer, early process controllers, speedometer.
Digital	Works on discrete values, represented as binary digits 0 and 1. It counts rather than measures, so results are exact to the chosen precision.	Almost all modern computers: PCs, phones, servers.
Hybrid	Combines analog measurement at the front end with digital processing behind it. Fast analog sensing feeds accurate digital computation.	Hospital intensive-care monitors, industrial process control, flight simulators.

5.2 By size and purpose

CLASS	TYPICAL CHARACTER	EXAMPLES AND USES

Microcomputer	Built around a microprocessor; designed for one user at a time; lowest cost of the four classes.	Desktop PC, laptop, tablet, smartphone; also embedded controllers in washing machines and looms.
Minicomputer	Mid-range machine supporting tens to hundreds of users at once; more storage and processing power than a microcomputer.	Departmental servers, factory control systems in medium-sized mills.
Mainframe	Very large memory and processing capacity; hundreds to thousands of simultaneous users; prized for reliability and huge transaction volumes.	Banking, airline reservation, and government census systems.
Supercomputer	The fastest class, built from thousands of processors working in parallel; performance measured in floating-point operations per second.	Weather forecasting, molecular modelling, fluid dynamics, cryptography research.

The boundaries blur over time — a modern smartphone exceeds a 1980s mainframe in raw speed — but the classification remains useful because it describes the intended scale of work, the number of users, and the cost.

6. Number systems and conversions

Computers store everything as binary digits because a two-state device (on or off, high or low voltage) is simple and reliable to build. People, however, work in decimal. Four number systems must be understood, each defined by its base and its allowed digits: **binary** (base 2, digits 0–1), **octal** (base 8, digits 0–7), **decimal** (base 10, digits 0–9), and **hexadecimal** (base 16, digits 0–9 and letters A–F for values 10–15). In any positional system, the value of a digit is the digit multiplied by the base raised to the power of its position, counting from zero at the rightmost digit.

$$\text{Value} = \Sigma\ \text{digit} \times \text{base}^{\text{position}}$$

WORKED EXAMPLES

Binary to decimal: $101101_2 = 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 = 32 + 0 + 8 + 4 + 0 + 1 = \mathbf{45_{10}}$.

Decimal to binary (repeated division by 2, read remainders upwards): 45 ÷ 2 = 22 r1; 22 ÷ 2 = 11 r0; 11 ÷ 2 = 5 r1; 5 ÷ 2 = 2 r1; 2 ÷ 2 = 1 r0; 1 ÷ 2 = 0 r1. Reading the remainders from last to first gives $\mathbf{45_{10} = 101101_2}$ — the reverse of the first calculation, as it must be.

Binary addition: 1011 + 0110. Units: 1 + 0 = 1. Twos: 1 + 1 = 10, write 0 carry 1. Fours: 0 + 1 +

carry 1 = 10, write 0 carry 1. Eights: 1 + 0 + carry 1 = 10, write 0 carry 1. Final carry becomes a fifth digit: **$1011 + 0110 = 10001_2$**. Check in decimal: 11 + 6 = 17, and $10001_2 = 16 + 1 = 17$. Correct.

Hexadecimal to decimal: $2F_{16} = 2\times16^1 + F\times16^0 = 32 + 15 = \mathbf{47_{10}}$. Hexadecimal is widely used as a compact shorthand for binary, because one hex digit is exactly four binary digits: $2F_{16} = 0010\ 1111_2$.

Octal works the same way with base 8, and one octal digit is exactly three binary digits. Grouping binary digits in threes (for octal) or fours (for hexadecimal) from the right is therefore a complete and exact conversion method in both directions, with no arithmetic needed.

7. Data representation and storage units

The smallest unit of data is the **bit** (binary digit), a single 0 or 1. Eight bits form one **byte**, enough to store one character such as a letter or a digit using a code such as ASCII. Larger units are defined in powers of 1024, because $1024 = 2^{10}$ is the power of two closest to one thousand:

UNIT	EQUALS	TYPICAL USE
1 byte (B)	8 bits	One character
1 kilobyte (KB)	1024 bytes	A short paragraph of text
1 megabyte (MB)	1024 KB = 1,048,576 bytes	A photograph or a short song
1 gigabyte (GB)	1024 MB	Part of a film; typical phone RAM sizes
1 terabyte (TB)	1024 GB	A large hard disk

WORKED EXAMPLE

Convert 2 GB into megabytes and kilobytes. 2 GB = 2 × 1024 MB = **2048 MB**. Then 2048 MB = 2048 × 1024 KB = **2,097,152 KB**. In bytes this would be 2,097,152 × 1024 = 2,147,483,648 bytes. Note the working rule: multiply by 1024 for each step down the ladder, divide by 1024 for each step up.

Text is stored as character codes, numbers as binary values, and images, audio, and video as long streams of bytes describing pixels, samples, or frames. Whatever the content, at the lowest level the machine sees only patterns of bits; meaning is supplied by the program that interprets the pattern.

8. Applications of computers

Computers are used in education (online classes, digital libraries, result processing), healthcare (patient records, imaging, diagnosis support), banking (accounts, ATMs, mobile banking), business (accounting, payroll, inventory), engineering and science (simulation, data analysis), entertainment, government, and defence. Two qualities explain this reach: any task that can be expressed as clear rules on data can be automated, and the cost of computing keeps falling.

In the textile industry the applications are direct and economically important. Computer-aided design (CAD) systems create and modify fabric and garment patterns far faster than hand drafting, and computer-aided manufacturing (CAM) drives cutting and stitching from those patterns. Automated looms and knitting machines run under computer control, with sensors reporting faults and production counts in real time. Dyeing and finishing lines use computers to control temperature, time, and chemical dosing, which improves shade consistency. Inventory and enterprise resource planning (ERP) software track yarn, fabric, orders, and shipments across a whole factory, and testing laboratories use computers to record tensile strength, colour fastness, and other quality results. A textile engineer who understands computing can use every one of these tools more effectively.

Glossary

TERM	MEANING
Program	A stored, ordered set of instructions that tells the computer how to process data.
ALU	Arithmetic and Logic Unit; the part of the CPU that performs calculations and comparisons.
Control unit	The part of the CPU that fetches, decodes, and coordinates the execution of instructions.
Peripheral	Any input or output device connected to the central system, such as a printer or scanner.
Bit / byte	A bit is a single binary digit (0 or 1); a byte is 8 bits, enough for one character.
Microprocessor	A complete CPU fabricated on a single chip; the defining technology of the fourth generation.
Hexadecimal	Base-16 number system using digits 0–9 and letters A–F; a compact shorthand for binary.
GIGO	Garbage In, Garbage Out: wrong input produces wrong output even on a perfect machine.

Key points & summary

- A computer accepts data, processes it under a stored program, and produces information; input, processing, output, and storage form one cycle.
- Its defining characteristics are speed, accuracy, diligence, versatility, large storage, and automation — balanced by a complete lack of independent intelligence.
- The CPU contains the ALU (calculates and compares), the control unit (fetches, decodes, coordinates), and the memory unit; input and output units connect it to the outside world.
- Five generations run from vacuum tubes through transistors and integrated circuits to the microprocessor and the present artificial intelligence era.
- Computers are classified by data handling (analog, digital, hybrid) and by size and purpose (micro, mini, mainframe, supercomputer).
- All data reduces to bits; 1 byte = 8 bits and each larger unit is 1024 of the previous one, so 2 GB = 2048 MB = 2,097,152 KB.
- Number conversions follow place value: $101101_2 = 45_{10}$ and $2F_{16} = 47_{10}$; binary addition carries just as decimal addition does.

Practice

Answer in your own words; brief model answers follow.

1. Explain, with one example each, the characteristics of diligence and versatility. Why do neither of them imply that a computer can think?
2. Convert 110010_2 to decimal, and convert 58_{10} to binary. Show your working for both.
3. Add the binary numbers 1101 and 1011, and check your answer by converting all three numbers to decimal.
4. A backup is 5 GB in size. How many megabytes is that, and how many kilobytes?
5. Distinguish between an analog, a digital, and a hybrid computer, giving one real application of each.
6. Describe two specific uses of computers in a textile factory and state the benefit of each.

Answer key

1. Diligence: the machine repeats an operation, such as checking every row of a payroll, millions of times without tiring or losing accuracy. Versatility: the same machine designs a pattern in the morning and prepares accounts in the afternoon, simply by running different programs. Neither implies thought, because every action follows instructions written in advance by a person; the machine originates nothing.

2. $110010_2 = 32 + 16 + 0 + 0 + 2 + 0 = 50_{10}$. For 58_{10}: 58 ÷ 2 = 29 r0; 29 ÷ 2 = 14 r1; 14 ÷ 2 = 7 r0; 7 ÷ 2 = 3 r1; 3 ÷ 2 = 1 r1; 1 ÷ 2 = 0 r1; reading remainders upwards gives 111010_2. Check: 32 + 16 + 8 + 0 + 2 + 0 = 58.
3. $1101 + 1011 = 11000_2$. In decimal: 13 + 11 = 24, and $11000_2 = 16 + 8 = 24$. Correct.
4. 5 GB = 5 × 1024 = 5120 MB; 5120 × 1024 = 5,242,880 KB.
5. Analog computers measure continuous quantities and give approximate results, for example an analog process controller following a continuously varying pressure. Digital computers count discrete binary values and give exact results, for example a PC preparing accounts. Hybrid computers combine analog sensing with digital processing, for example an intensive-care monitor that senses continuous body signals and processes them digitally.
6. Any two of: CAD pattern design, which shortens design time and makes modification easy; computer control of automated looms, which raises speed and reports faults at once; ERP inventory tracking, which prevents stock-outs and over-ordering; computer-controlled dyeing, which improves shade consistency by holding time, temperature, and dosing steady.

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